Animal Kingdom — Comprehensive Notes

PU-I Biology • Full Chapter Revision

1. Basis of Classification

Over a million species of animals have been described so far, making classification essential. Basis of classification includes: levels of organisation, patterns of digestive/circulatory/reproductive systems, body symmetry, arrangement of cells in germ layers, nature of coelom, segmentation, and presence or absence of notochord.; levels of organisation: cellular level (e.g. Porifera), tissue level (e.g. Coelenterata), organ level (e.g. Platyhelminthes), and organ system level (e.g. Echinoderms and Chordates). Organ system pattern may be incomplete (one opening, e.g. Platyhelminthes) or complete (separate mouth and anus, with open or closed circulation).

2. Body Symmetry, Germ Layers and Coelom

radial symmetry: any plane through the central axis divides the body into identical halves (e.g. coelenterates, ctenophores, echinoderms). Bilateral symmetry: only one plane divides the body into identical left and right halves (e.g. insects, chordates).; Asymmetry: no plane divides the body into equal halves (e.g. sponges/Porifera). Diploblastic animals have ectoderm and endoderm with mesoglea in between (e.g. coelenterates); triploblastic animals have ectoderm, mesoderm and endoderm (e.g. Platyhelminthes to Chordates). Coelomates possess a mesoderm-lined body cavity (coelom) — e.g. Annelida, Mollusca, Arthropoda, Echinodermata, Hemichordata, Chordata.

3. Segmentation and Notochord

Segmentation: body externally and internally divided into segments with repeated organs, called metameric segmentation or metamerism (e.g. Earthworm). Notochord is a mesoderm-derived rod-like structure formed dorsally during embryonic development.; Animals with notochord are chordates; animals without notochord are non-chordates (e.g. Porifera to Echinodermata).

4. Phylum Porifera

Commonly called sponges; generally marine, primitive, multicellular, asymmetrical, with cellular level of organisation. Have a water transport / canal system: water enters through ostia into the spongocoel and leaves through the osculum, helping in food gathering, gas exchange and excretion.; Spongocoel and canals are lined by collar cells (choanocytes); digestion is intracellular. Skeleton consists of spicules or spongin fibres.

5. Phylum Coelenterata (Cnidaria)

Coelenterates/Cnidaria are aquatic (mostly marine), sessile or free-swimming, radially symmetrical, diploblastic, with tissue level of organisation. Have a central gastro-vascular cavity (GVC) with a single opening, the hypostome, surrounded by tentacles.; Tentacles have cnidoblasts/cnidocytes containing nematocysts, used for anchorage, defence and capturing prey — the name Cnidaria comes from cnidoblasts. Digestion can be extracellular and intracellular; skeleton (in corals) is composed of calcium carbonate. Two basic body forms: polyp (sessile, cylindrical, e.g. Hydra, Adamsia) and medusa (umbrella-shaped, free-swimming, e.g. Aurelia/jellyfish).

6. Phylum Ctenophora

Commonly known as sea walnuts, comb jellies or sea gooseberries; exclusively marine, radially symmetrical, diploblastic, tissue level of organisation. Digestion is extracellular or intracellular; locomotion is by eight rows of externally located ciliated comb plates.; Sexes are not separate (hermaphrodite); reproduction is sexual only with external fertilisation and indirect development (cydippid larva).

7. Phylum Platyhelminthes

Commonly called flatworms due to dorso-ventrally flattened bodies; bilaterally symmetrical, triploblastic, acoelomate, organ level of organisation. Nutrition is parasitic (endoparasites) with hooks and suckers present, or nutrients absorbed directly from the host.; Excretion through specialised flame cells/solenocytes/protonephridia, which aid osmoregulation and excretion. Reproduction sexual and asexual (fission); Platyhelminthes are hermaphrodites with internal fertilisation and development consisting of several larval stages; some show high regeneration capacity. Examples: Taenia (tapeworm, parasite in pigs and humans), Fasciola (liver fluke, parasite in goat/sheep), Dugesia (Planaria, aquatic free-living).

8. Phylum Aschelminthes

Known as round worms due to circular cross-section body; bilaterally symmetrical, triploblastic, pseudocoelomate, organ system level of organisation. May be free-living (aquatic/terrestrial) or parasitic on plants and animals; alimentary canal is complete with a well-developed muscular pharynx.; Excretory tube removes waste through the excretory pore. Aschelminthes are dioecious (sexes separate), with females often longer than males; fertilisation is internal, development direct or indirect. Examples: Ascaris (round worm, causes Ascariasis), Wuchereria (filarial worm, causes Filariasis/Elephantiasis transmitted by mosquitoes — Anopheles in Africa, Culex in America), Ancylostoma (hookworm, infects small intestine of cats and dogs).

9. Phylum Annelida

May be aquatic (marine/freshwater) or terrestrial; bilaterally symmetrical, triploblastic, coelomate, organ system level, with metamerically segmented bodies. Locomotion by longitudinal and circular muscles; parapodia present in aquatic forms for swimming.; Circulatory system closed; excretory organs are nephridia; nervous system has paired ganglia connected to a double ventral nerve cord by lateral nerves. Reproduction is strictly sexual; some forms dioecious (e.g. Nereis) while some are monoecious/hermaphrodite (e.g. Earthworm and Leech). Examples: Nereis (clam worm), Pheretima (earthworm), Hirudinaria (leech).

10. Phylum Arthropoda

largest phylum in Animalia, comprising over two-thirds of known animal species, mostly insects; bilaterally symmetrical, triploblastic, coelomate (haemocoel — coelom filled with blood). Body covered by chitinous exoskeleton with distinct head, thorax and abdomen; have jointed appendages. Respiration through gills/book gills (aquatic) or book lungs/tracheal system (terrestrial); circulatory system is open; excretory organs are Malpighian tubules.; Sensory organs include compound and simple eyes, antennae and statocysts (balance organs). Mostly dioecious with internal fertilisation and direct or indirect development. Examples: Apis (honey bee), Bombyx (silkworm), Laccifer (lac insect); vectors like Aedes, Anopheles, Culex; pests like Locusta and cockroach; Limulus (king crab) is a living fossil.

11. Phylum Mollusca

second largest phylum in Animalia among invertebrates; terrestrial or aquatic; bilaterally symmetrical, triploblastic, coelomate, organ system level. Body covered by a calcareous shell with distinct head, muscular foot and visceral hump; a soft mantle covers the visceral hump. Mantle cavity (space between hump and mantle) contains feather-like gills with respiratory and excretory functions.; Excretory organs: one or two pairs of kidneys called metanephridia; sensory tentacles on the head; feeding organ is the file-like radula. Molluscs are dioecious and oviparous, with direct or indirect development. Examples: Pila (apple snail), Pinctada (pearl oyster), Octopus (devil fish), Sepia (cuttlefish), Loligo (squid), Aplysia (sea-hare), Dentalium (tusk shell), Chaetopleura (chiton).

12. Phylum Echinodermata

Echinodermata means spiny-bodied due to an endoskeleton of calcareous ossicles; exclusively marine; bilaterally symmetrical, triploblastic, coelomate, organ system level. Digestive system is complete with mouth on the ventral side and anus on the dorsal side.; Water vascular / ambulacral system is a distinctive feature, helping in locomotion, food capture/transport and respiration. Reproduction is sexual with external fertilisation and indirect development (Bipinnaria larva). Examples: Asterias (star fish), Echinus (sea urchin), Cucumaria (sea cucumber), Ophiura (brittle star), Antedon (sea lily).

13. Phylum Hemichordata

Earlier placed as a sub-phylum of Chordata; a small group of worm-like marine animals; bilaterally symmetrical, triploblastic, coelomate, organ system level. Body is cylindrical, composed of anterior proboscis, collar and a long trunk.; Circulatory system open; respiratory organs are gills; excretory organ is the proboscis gland. Sexes separate, fertilisation external, development indirect. Examples: Balanoglossus (tongue worm/acorn worm) and Saccoglossus.

14. Phylum Chordata — General Features and Subphyla

Fundamental characters of Chordata: presence of notochord, dorsal hollow nerve cord, and paired pharyngeal gill slits. Bilaterally symmetrical, triploblastic, coelomate, organ system level; possess a post-anal tail and a closed circulatory system.; Phylum Chordata is divided into three subphyla: Urochordata/Tunicata (e.g. Ascidia, Salpa, Doliolum, Herdmania), Cephalochordata (e.g. Branchiostoma/Amphioxus or Lancelet), and Vertebrata/Craniata.

15. Class Cyclostomata

Sucking circular mouth without jaws; elongated, eel-like body. Marine but migrate to fresh water for spawning; die shortly after spawning; larvae metamorphose and return to the ocean.; Ectoparasitic on other fishes; respiration by 6-15 pairs of gill slits; body devoid of scales and paired fins; cranium and vertebral column are cartilaginous. Examples: Petromyzon (lamprey), Myxine (hag fish).

16. Class Chondrichthyes

Marine habitat; cartilaginous endoskeleton, streamlined body; notochord persists throughout life. Skin has small placoid scales; predaceous with strong jaws, mouth located ventrally, teeth are modified placoid scales directed backward. Respiration by 5-7 pairs of gills without operculum; must swim continuously to avoid sinking (no air bladder); heart two-chambered (1 auricle, 1 ventricle).; Some have electric organs (e.g. Torpedo, 8-220 volts) or poisonous stings (e.g. Trygon/sting ray); poikilothermous. Sexes separate; males have claspers on pelvic fins; internal fertilisation; many are viviparous. Examples: Scoliodon (dog fish), Pristis (saw fish), Carcharodon (great white shark), Trygon (sting ray).

17. Class Osteichthyes

Marine and freshwater habitats; bony endoskeleton, streamlined body with terminal mouth. Body covered by cycloid or ctenoid scales; respiration by four pairs of gills covered by operculum; air bladder provides buoyancy.; Heart two-chambered; poikilothermous; sexes separate, external fertilisation, mostly oviparous with direct development. Examples: Marine — Exocoetus (flying fish), Hippocampus (sea horse); Freshwater — Labeo (rohu), Catla, Clarias (magur); Aquarium — Betta (fighting fish), Pterophyllum (angel fish).

18. Class Amphibia

Can live both in water and on land; body divisible into head and trunk with two pairs of limbs; tail may be present in some. Alimentary canal, excretory and reproductive tracts open into a common cloaca; skin is moist without scales. Eyes have eyelids; ear represented by tympanum; respiration by gills, lungs and skin.; Heart three-chambered (two auricles, one ventricle); poikilothermous. Sexes separate, external fertilisation, oviparous, indirect development. Examples: Bufo (toad), Rana (frog), Salamandra (salamander), Hyla (tree frog), Ichthyopis (limbless amphibian).

19. Class Reptilia

Characterised by creeping/crawling locomotion; mostly terrestrial; body covered by dry cornified skin with epidermal scales/scutes. No external ears but tympanum present; two pairs of limbs when present; heart three-chambered (four-chambered in crocodiles); poikilothermous.; Snakes and lizards shed their scales as skin cast. Sexes separate; internal fertilisation; oviparous; direct development. Examples: Chelone (turtle), Testudo (tortoise), Chameleon, Calotes (garden lizard), Crocodilus, Alligator, Hemidactylus (wall lizard); poisonous snakes — Naja (cobra), Bangarus (krait), Vipera (viper).

20. Class Aves

Characterised by wings and flight ability (except flightless birds like ostrich); have beaks; forelimbs modified into wings, hind limbs used for walking/swimming/clasping branches. Skin is dry with oil glands only at the base of the tail; endoskeleton ossified with hollow, pneumatic long bones.; Respiration by lungs and air sacs; digestive system has crop and gizzard as additional chambers; heart four-chambered. Homoiothermous (warm-blooded); internal fertilisation, oviparous, direct development, sexes separate. Examples: Corvus (crow), Columba (pigeon), Psittacula (parrot), Struthio (ostrich), Pavo (peacock), Aptenodytes (penguin), Neophron (vulture).

21. Class Mammalia

Found in a wide variety of habitats — polar ice, deserts, mountains, forests, grasslands, dark caves; adapted to fly, swim or walk. Presence of mammary glands (producing milk for young) is a unique mammalian character; respiration by lungs.; Two pairs of limbs adapted for walking, running, climbing, burrowing, swimming or flying; skin covered by hair; external ears (pinnae) present. Heart four-chambered; homoiothermous; sexes separate; internal fertilisation; mostly viviparous (exception: Platypus is oviparous); direct development. Examples: Ornithorhynchus (platypus, oviparous), Macropus (kangaroo), Pteropus (flying fox), Camelus (camel), Macaca (monkey), Rattus (rat), Canis (dog), Felis (cat), Elephas (elephant), Equus (horse), Delphinus (dolphin), Balaenoptera (blue whale), Panthera tigris (tiger), Panthera leo (lion).

22. Summary — Salient Features of Phyla

This summary table brings together level of organisation, symmetry, coelom, segmentation, digestive/circulatory/respiratory systems, and distinctive features for all eleven phyla covered in this chapter — ideal for quick pre-exam revision.; Use the salient comparisons in this topic to distinguish groups accurately in classification questions.